INCH-POUND

MIL-I-6301B
15 SEPTEMBER1994
SUPERSEDING
MIL-I-6301A
29 February 1952

MILITARY SPECIFICATION

INDICATOR, PRESSURE, FUEL, O-600 PSI

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hermetically sealed, remote indicating, synchro-style, fuel pressure indicator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/ TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-6032	Transmitter, Pressure, Fuel, 0-600 PSI
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For

STANDARDS

MILITARY

MS28005	Transmitters, Pressure, Synchro, Aircraft
MS28010	Indicators, Pressure - Synchro, Single, 2-Inch Size
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. Department Of Labor, Occupation, Safety, And Health Administration (OSHA)
29 CRF 1910 - Occupation, Safety, And Health Standards

(Copies of 29 CFR 1910 are available from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402-0001.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D4727 Corrugated And Solid Fiberboard, Sheet Stock (Container Grade) And Cut Shapes

(Application for copies should be addressed to: ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government Standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.1.2 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 6.3).

3.2 Design. The indicator shall be designed for use with a MS28005-7 fuel pressure transmitter in accordance with MIL-T-6302.

3.3 Construction.

3.3.1 Case. The case shall be hermetically sealed and the case dimensions shall conform to MS28010.

3.3.1.1 A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.3.1.2 The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed .031 inch.

3.3.1.3 Case finish. The portion of the case visible when mounted in the normal instrument panel shall be finished in durable dull black.

3.3.2 Pointer. The pointer shall be in accordance with MS33585-8, except that the length of the pointer shall be such that the tip will extend into the scale a distance equal to 0.33 to 0.67 of the length of the shortest graduation.

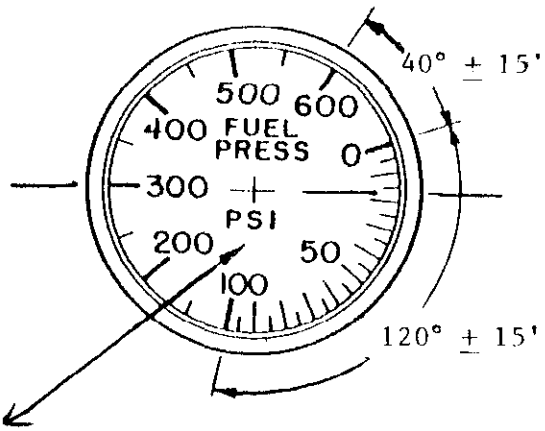
3.3.2.1 Pointer finish. The shaded portion of the pointer shall be finished with fluorescent-luminescent material.

3.3.3 Dial. The dial shall be as shown on figure 1. A minimum diameter of 1.75 inches shall be maintained across the outside ends of the graduations. The graduation for 300 psi shall be located on the horizontal center line at the left side of the dial. The electrical zero reference mark shall be located 160 degrees clockwise from the zero graduation.

3.3.3.1 Dial markings.

3.3.3.1.1 Fluorescent-luminescent markings. The markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

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MS28010-7
LOCATION OPTIONAL

DIMENSIONS IN INCHES

FIGURE 1. Dial

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MARKINGS	HEIGHT OR LENGTH INCH + 0.016	WIDTH OF LINE OR GRADUATION INCH + 0.005
Numerals 0, 50, 100, 300, and 500	0.156	0.025
Graduations at 0, 50, and each 100	0.188	0.031
Each 10-pound graduation 10 to 90	0.188	0.020
Five-pound graduations and graduations 150, 250, 350, 450, and 550	0.094	0.020
Lettering "FUEL PRESS" 0.094	0.025	
Lettering "PSI" 0.094	0.020	

3.3.1.2 Durable dull black markings. The markings MS28005-7 shall be permanently and legibly marked on the dial in 0.063 inch letters. These markings and all other markings not otherwise specified shall be finished in durable dull black.

3.3.4 Weight. The weight of the indicator shall not exceed 0.60 pound.

3.4 Identification of product. The nameplate shall be marked with the following information:

INDICATOR, PRESSURE, FUEL, 0-600 PSI
MS Part Number
Stock Number
Manufacturer's Part Number
Manufacturer's Serial Number
Contract or Order Number
Manufacturer's Name or Trademark
U.S. Property

3.5 Reclaimed materials. The indicator may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the indicator. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, remanufactured end products are allowed under this specification.

3.6 Safety and health requirements. All parts, components, and assemblies furnished on the indicator, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence

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of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 General. The fuel pressure indicator shall be subjected to inspection and tests in accordance with MIL-I-7057.

4.3 Test tolerances. The indicator shall be within the tolerances specified in Table I and II of this specification when tested in accordance with MIL-I-7057.

TABLE I

Pressure Indicator Tolerances ± psi

TEST PSI	ROOM TEMPERATUR	LOW TEMPERATURE	HIGH TEMPERATURE
0	1.0	1.0	1.0
50	1.0	1.0	1.0
100	1.0	1.0	1.0
150	3.0	3.0	4.0
200	3.0	3.0	4.0
300	3.0	3.0	4.0
400	3.0	3.0	4.0
500	3.0	3.0	4.0
600	3.0	3.0	4.0

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TABLE II

Tolerance PPH ±

TEST	0 - 100 psi	Over 100 psi
Position Error at 200 psi	1.0	2.0
Friction:		
Room Temperature	1.0	6.0
High Temperature	1.0	6.0
Low Temperature	1.5	9.0
Vibration:		
Pointer Oscillation	0.5	3.0
Pointer Variation	0.5	3.0

4.4 Inspection and test. Test of methods of preservations and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to ensure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each pressure indicator shall be packaged in quantity unit packs of one each in accordance with Method IA8 of MIL-P-116. Each pressure indicator shall be packed in fiberboard shipping containers conforming to PPP-B-636, Box Shipping, Fiberboard, or ASTM D4727, Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes. Sufficient cushioning material between the bag and unit container shall be provided. The cushioning material shall be of the proper type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subject to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in fiberboard shipping containers conforming to PPP-B-636 or ASTM D4727, unless otherwise specified by the contracting activity. When practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required. All other requirements, as specified by the contracting activity, shall apply.

5.2.2 Level B. Indicator packaged as specified in 5.1.1, shall be packed in fiberboard shipping boxes conforming to PPP-B-636 or ASTM D4727, unless otherwise specified by the contracting activity. All other requirements, as specified by the contracting activity, shall apply.

5.2.3 Level C. Packaging shall be applied which affords protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Package marking. Each interior package shall be marked with the following information:

INDICATOR, PRESSURE, FUEL, SYNCHRO, 2-INCH SIZE, 0-600 PSI
26 Volts 400 Cycles
MS Part Number
Stock Number
Manufacturer's Part Number
Manufacturer's Serial Number
Contract or Order Number
Date of Manufacture
Quantity
Name of Manufacturer
Name of Contractor (if different from manufacturer)

IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE USE.

*This date shall be 36 months from the date of inspection and shall be inserted by the inspector at the time of acceptance.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fuel pressure indicator covered by this specification is intended for use in conjunction with a suitable transmitter in indicating remotely the pressure in the fuel systems of aircraft engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation, packaging, and packing required.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key words) listing. Recommended key words are:

Fuel Pressure Indicator
Hermetically sealed
Synchro Style

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99
Navy - AS

Preparing activity:
Air Force - 71

Project number:
6620 - 0532

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